INTEGRATING FLOW THEORY IN FREIREAN PEDAGOGY BASED WORKSHOPS FOR FORMATIVE ASSESSMENT

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Abstract

This work-in-progress paper discusses, in context of our workshops for programming data analytics APIs, the student disengagement problems. The problems arise during designing the workshops to enable culturally responsive **F**ormative **A**ssessment (hereafter FA) for freshmen students with different culture backgrounds. The workshops are organized as informal and optional offers in context of our university STEM courses. In our culturally responsive actions for FA, our main difficulty is to ensure that the challenges in goal-definition is calibrated to student's available skills and student's needs (e.g. psychological needs) at cultural or social contexts. While goals that are very high challenging lead to a student's anxiety, goals that are too easy generate student's boredom. Being in psychological states of boredom or anxiety increases the possibility of disengagement phenomena. To investigate and better understand the disengagement phenomena in different cultural oppressive structures, the paper suggests a practice-method conceptualized as helping tool for trainers. It helps trainers practice how to use and integrate Csikszentmihalyi's flow-model [1] and Freirean Dialogues for FA. it enables three responsive functions for FA: namely 1) function for flow-model based diagnosing of student's emotional states (e.g. boredom, apathy, worry, and anxiety) that hinder his/her engagement, 2) function for analysing student's oppressive cultural structures/lived situations that hinder his/her engagement, and 3) function for assessing student's available cognitive skills to facilitate matching balance between challenges and skills in goal-definition. To put it differently, we have integrated the flow-model as goal-codification form in Freirean dialogue for action and reflection. In this form, the trainers can codify the challenges and skills of a goal in a measurable style for better analysing and understanding the disengagement problems. Finally, we present how trainers and students can perceive our practice-method as inquiry process for co-humanists who are interested in investigating the act of knowing as act for liberation and reclaiming full humanity.

Keywords: Csikszentmihalyi's Flow theory, humanist psychology, Freire's pedagogy of the oppressed, humanist pedagogy, formative assessment and measurement, assessment in engineering education, data analytics.

1 LITERATURE REVIEW AND THE CONTRIBUTIONS OF OUR RESEARCH BASED ARCHITECTING PROJECT

Assessment is usually understood to have two purposes: summative and formative. Summative assessment is used to judge students' achievements at the end of an instructional unit, whereas formative assessment aims to gather evidence about students' learning progress to influence teaching methods and priorities. According to Hattie [2], using two dividing notions (i.e. formative and summative) is not helpful in education debate about assessment. To improve the debate Hattie suggests to use, instead of two dividing notions, the single notion "Assessment-capable learner". This notion enables him to argue that both forms of assessments: summative and formative are important [2]. To support his argument, he used the metaphor suggested by Robert Stake [3] that draws not only a clear distinction line between formative and summative assessments but also highlights the importance of them both. The metaphor is: "*When the cook tastes the soup, that's formative and when the guests taste the soup, that's summative*". That means: for the task of a teacher as evaluator and the task of the student as self-evaluator; both evaluating the paths during learning (i.e. formative assessments) and evaluating a goal achievement in a learning path (summative assessments) are helpful and important. In stake's metaphor, tasting the soup by cook during making it and by guests during serving it is helpful and important. Hattie's Visible Learning pedagogy was developed in context of primary schooling, therefor our investigations in his pedagogy, during our deductive based research activities in [4]), aim to test if his pedagogy and its explaining constructs hold or do not hold in contexts of our university offers for freshman students with different cultural backgrounds. The offers are for teaching and training software design and modern programing languages. In context of these offers, understanding the pedagogy-

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assessment linkage is the main objective of our research as software architects. In our research based architecting activities, we formulate our research questions (e.g. questions in [4,5,6]) to practice two types of research; namely deductive and inductive. In deductive research, we start from one or more theories in pedagogy and/or psychology (e.g. Freire's pedagogy of the oppressed [7], and Csikszentmihalyi's Flow theory [1]), whereas in inductive research we start from phenomena observations (e.g. observations for learned helplessness phenomena or for student disengagement phenomena).

The educational philosophies of John Dewey [8], Paulo Freire [9], and other advocates of progressive and critical pedagogy can be regarded, according to Noam Chomsky [10], as further developments of Humbold's philosophy for educational reforms started in Germany about 200 years ago. In terms of educational contexts, present-day German universities may seem far removed from the challenges and concerns that troubled Freire when he began to develop his ideas of liberatory education and pedagogy of the oppressed. However, we are interested in investigating if Freire's dialogic pedagogy and its explaining constructs hold or do not hold in context of some offers provided by us for freshman students, and for students with immigration background because of the wars in Europe (e.g. war in Ukraine) and in Middle East (e.g. war in Syria).

Our research based architecting activity in [5] was about the FA implementation challenges in context of project based pedagogy for students in third semester of three years based computer science program, whereas in [6] the research based activity was about the FA implementation challenges in course based pedagogy context. The context was for a teacher of Computer Organization (CO) course that has about 130 freshman students. In contrast to [5, and 6], in this paper we are interested in the FA implementation challenges in context of our purpose to develop literacy program of workshops, using Freire's pedagogy of the oppressed, for students with different culture backgrounds (e.g. students immigrated from Ukraine or Syria because of the war)

Our main contribution in this paper is:

- We have conceptualized a practice-method as helping tool for trainers. It helps trainers practice how to use and integrate Csikszentmihalyi's flow-model [1] and Freirean Dialogues for FA. it enables three culturally responsive functions for FA: namely 1) function for flow-model based diagnosing of student's emotional states (e.g. boredom, apathy, worry, and anxiety) that hinder his/her engagement, 2) function for analysing student's oppressive cultural structures/lived situations that hinder his/her engagement, and 3) function for assessing student's available cognitive skills to facilitate matching balance between challenges and skills in goal-definition. Moreover, the responsive functions for FA facilitate the liberation-oriented defining and assessing of student self-goals (sections 3 and 4).
- We have integrated the flow-model as goal-codification form. In this form, the trainers can codify the challenges and skills of a goal in a measurable style for better analysing and understanding the disengagement problems (section 4.2).
- We present how trainers and students can perceive our practice-method as inquiry process for co-humanists who are interested in investigating the act of knowing as act for reclaiming full humanity (section 4.3)

The main contribution in this paper is an example for a deductive research activity's outcomes during our research based architecting iterations. The activity's inputs, outcomes and main steps are briefly summarized in next section.

2 METHOD: ARCHITECTING BASED ON DEDUCTIVE RESEARCH

Fig 1. Summarizes the method we use for architecting based on deductive research. It is deductive because we start in investigation from a theory. It is a feedback-loop of two acts: act for critically perceiving and act for proof of value.

- Act for critically perceiving: In this act we use our systems thinking skills as software architects to perceive critically a theory or theoretical view as an explaining system that might help us better understand an aspect of the phenomenon (e.g. disengagement) under study or analysis. The act aims to conduct a critical interpretation or critical analysis for trade-off decision making or for testing (e.g. testing if an explaining construct of a theory holds or not at a social or cultural context). Perceiving a theory as an explaining system (i.e. a set of explaining constructs), enable us also to study the integration of two explaining systems. For example, in this paper our goal is to study and critically analyse how the trainers can use and integrate the flow theory and Freire's pedagogy to better understand the disengagement problem.

- Act for proof of value: this act requires as input an interpretation or/and test outcome. It produces as outcome a value based architecture as help tool (see section 4 as an example).

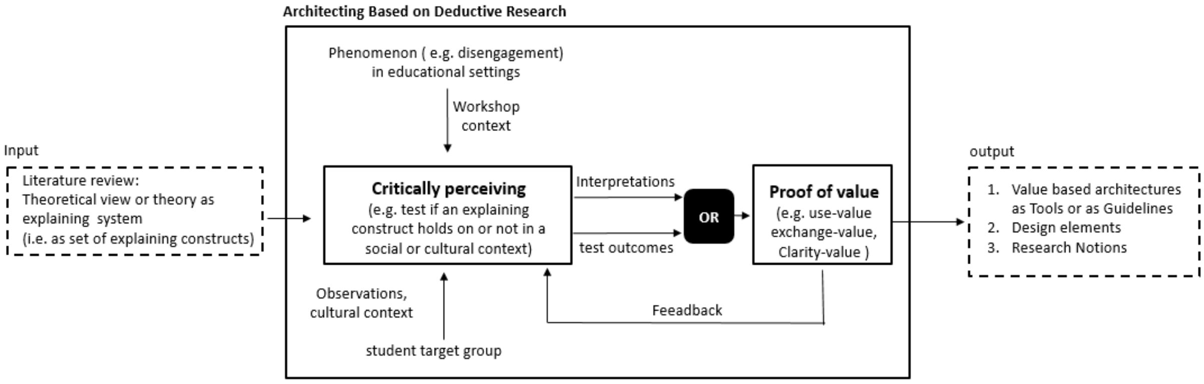

Figure 1. Architecting based on deductive research

Moreover, for better systematic organized work, we plan to organize and conduct our design tasks at three different levels; **reflective, behaviour and visceral (**defined by Don Norman [11]**)**. For these tasks the deductive research in Fig .1 is not required, therefor we use, to conduct them, two powerful and famous tools (see Fig 2.1 and 2.2) in design thinking literature, namely; double-diamond diverge-converge model, and the iterative cycle of human-centered design [11, p 220-225]

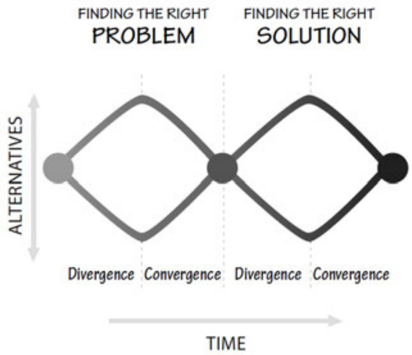

Figure 2.1 double-diamond diverge-converge model (Slightly modified by Don Norman from the work of the British Design Council, 2005.) [11, p 220-225]

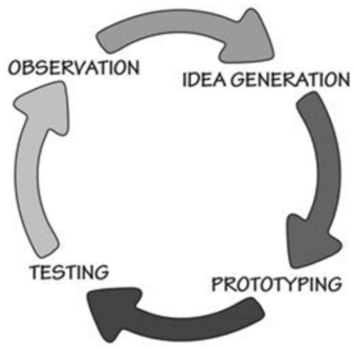

Figure 2.2 The Iterative Cycle of Human-Centered Design) [11, p 220-225]

To sum up, driven by a value (e.g. use-value or clarity-value) for deductive research task, our method as feedback-loop of two acts enables conducting a serie of critical interpretations or analyses. For this paper the serie's initial critical analysis requires inputs presented in next section.

3 INPUTS FOR INITIAL CRITICAL ANALYSIS

We have two inputs; The first is the workshop context (i.e. our goal-definition practices) in which the disengagement phenomenon is observed, and the second is the target groups of students we are interested in better understanding of their disengagement and its root causes.

3.1 Workshop context: Our Goal-Definition Practices

During our workshops, we collect feedback to reflect on our goal-definition practices for refinements. The refinements are summarized by us in usable input format for further critical analysis tasks. For initial analysis tasks in this paper, we present two refinements: namely refinement for goal type, and refinement for goal-definition form.

Refinement for goal type: the goal-definition practice in our workshops must differentiate between the types of student's self-goals: There is a rich literature on the self-goals that students can have. According to Hattie [12], there are three types of goals described in table (1). For clarifying our focus on mastery goals and not on performance or social goals, the differences between social goals, performance goals and mastery goals are shared with students who participate in our workshops.

Table 1. Types of Student's self-goals.

Type	*Description*
Mastery	Mastery goals arise when students aim to develop their competences (i.e. skills) and they consider ability to be something that can be developed by increasing effort.
Performance	Performance goals arise when students aim to demonstrate their competence particularly by outperforming peers and they consider ability
Social	Social goals arise when students are concerned about how they interact with, and relate to, others in the class.

Refinement for goal-definition form: the goal-definition practice must include a definition form for goal assessability/measurability. A measurable/assessable goal is required for supporting a student in matching balance between his/her progressive challenges and his/her available skills in relation to his/her emotional state. The student's available cognitive skills and emotional states are modeled using two separate dimensions for measurability/assessability, namely; dimension for student's cognitive skills and dimension for student's emotional states. Moreover, the forms of goal-definition under study are so many in context of our workshops and they are documented for further analysis in problem space in two styles; detailed and briefing. Using the briefing style, table 2 summarize the main six concrete examples of goal-definition forms emerged during developing a program of workshops for responding to students' changing needs in programming APIs.

Table 2 Forms of goal-definition briefly documented for further analysis in problem space

ID	*Title*	*Description*
1	**Form** for **E**xternal **D**esign (hereafter **ED Form**) of an API coding challenge.	This is about expected outcomes when a challenge in a goal is achieved. The focus here is not how to implement a solution but what a solution's outcomes must be. The solution's outcomes are defined as a measurable outcome state. This state definition is independent of the programing or implementation languages.
2	**Form** for **I**nternal or **I**mplementation **D**esign (hereafter **ID Form**) of an API coding challenge.	This is about how to implement a solution for a challenge. To put it differently, it is about the solution's implementation or programming styles for a challenge. It depends on the language used for implementation and it can be replaced easily by other implementations to enable quality assurance in design for incremental challenges.
3	**Form** to enable API **Q**uality **A**ssurance Metrics in **C**ontainer **D**esign (hereafter **QA CD** Form)	The outcome of this definition-form is a QA container that facilitates quality assurance metrics in incremental challenge design for skills mastery improvement in API programing. The QA container has student challenges that are not only clear and actionable but also easy to be selected and combined by the student in context of activating him/her as owner of his/her learning.
4	**C**ulture **B**ased **P**resentation **Form** (hereafter **CBP** Form) for solving communicating problems in some cultures.	This form helps the trainers to understand the student's changing needs in different cultures. The two culture settings; indirect and direct are defined to analyze and understand some communicating problems with students. By "indirect" and "direct", we mean: 1. Indirect expressing style used by a student because he/she was educated by family or by primary schooling to avoid expressing the point or his/her needs. 2. Direct expressing style. This is because the student was educated to express directly the point or his/her needs.

5	Form to enable anonymous comments based feedback for student's engagement	Description: This form is constructed by us to solve a disengagement problem. The problem is: Our review feedback as comments on a student's request for finalizing a goal's challenge (i.e. a Merge Request [13]) were not effective because most of the students cannot get the reviewer's point of view and they do not invest time to read carefully the review comments. (They are in a disengagement state). The solution for this problem was to provide ***anonymously*** previous reviews and their comments, related to students in last year, beside a goal specification. And then ask the student to find out what review and comments might be for the goal at hand. In this solution, the probability of student engagement to understand the reviewer's point of view and to understand the success criteria of a goal (i.e. the knowledge of the goal's endpoints) is increased exponentially.
6	Form for anonymous short-reports at workshop exit.	This form enables the trainer to effectively planning the starting instruction points in next workshop iteration based on learning outcomes extracted from analyzing the self-reports at workshop exit.

The form 5 's documentation in table 2, introduces a problem, in written communication context, that hinders students to engage with trainer's review comments. It also suggests an effective solution via delivering a feedback in form of anonymous comments for facilitating student's engagement. In contrast to form 5's problem in written communication context, the context of the problems reported by form 4 (see row 4 in table 2) is in oral and face-to-face communication. The communication problems arise during our try to clarify goals in different cultural contexts. To analyze the problems and its culture contexts, we initially classify them for further study in two classes of culture settings; indirect-culture and direct-culture. Moreover, to facilitate our systematic analysis of goal-communication problems, a class of problems in indirect-culture, and a class of problems in direct-culture are formulated for our further study and analysis. Concrete examples for analyzing problems in indirect-culture and in direct-culture can be found in our work [4]. In this paper, we focus only on goal-communication problems with group of students in indirect-culture. In next section, we introduce the student target groups.

3.2 Target Student Groups in Indirect-Culture

By "indirect" in "indirect-culture" we refer to indirect style in communication and in expression used by a student. The indirect style is used by him/her because he/she suffers from cultural oppressive structures during his/her pre-university schooling. To put it differently, he/she was emerged as cultural product from a culture of silence in his/her society. Moreover, due to the society's oppressive structures he/she had no rights to express his authentic and individual voice. Moreover, he/she had no kid rights as the rights of kids in modern and humanitarian societies. For example, he/ she has forced to work as kid for economic reasons. For this paper, we limit our investigation to understand the oppressive structures in linguistic phenomena. The linguistic oppressive structures in pre-university schooling for adult students from Syria, Egypt, and Jordan were planned to be investigated. For this investigation, collecting data of oppressive linguistic structures will be conducted by observation methods from linguistic anthropology. In linguistic anthropology [14], we approach language as culture resource and speaking as cultural practice. The analysis of initial observations for adult students from Syria, Egypt, and Jordan is not planned to be reported in this paper but in future separate paper.

Not only in pedagogical dimension but also in phycological dimension we plan to investigate and analysis the disengagement phenomena. To help trainers use and integrate the flow theory in psychology and Freire's pedagogy in their tasks to investigate and better underrated the students' disengagement, we present and discuss the results of our research based architecting in next section.

4 RESULTS AND DISCUSSION

The results and their discussion are structured in three parts; the first is about our practice-method's method architecture as guideline to integrate Freirean dialogues and flow-model, the second is about how to use the flow-model to codify the challenges and skills of a goal in a measurable style to better analyse and understand student's disengagements, and the third is about a co-humanist role designed to clarify how the method can be used for what groups of oppressed students.

4.1 Our Practice-method's Architecture as Guidelines

Fig. 3 shows, in big picture, our method's architecture as guidelines for integrating Freirean dialogues and flow model in initializing a program of workshops for a group of oppressed students.

Given a group of oppressed students who are interested in education act for liberation, our method's architecture is conceptualized as tools to help trainer and students in systematic co-inquiry and co-design of adult literacy program of workshops. The method conceptualizes systematically the co-inquiry and co-design tasks by using and integrating Freire's pedagogy and Flow theory [1]. In the first version of this method, we present an architecture and its design elements as tools to help co-humanists (i.e. trainers and students using Freirean styles of inquiry) to use and integrate Freirean dialogues and Csikszentmihalyi's Flow-model in initializing a program of workshops as act for liberation. Fig.3 shows guidelines to enable three **responsive functions** for FA. The functions are:

1 Analysing and investigating, via freirean dialogue for goal-definition, the student's oppressive cultural and linguistic structures/lived situations that hinder his/her engagement.

2 Assessing and diagnosing, via flow-model based freirean dialogue for reflection, student's emotional states (e.g. Boredom, apathy, worry, and anxiety) that hinder his/her engagement.

3 Assessing, via freirean dialogue for reflection, student's available cognitive skills to facilitate matching balance between challenges and skills in goal-definition

The three responsive functions for FA as organized and structured in Fig.3 are our initial architectural interpretation after reading Freire's pedagogy in [7,15,16] and Flow theory in [1,17]. Our initial interpretation is the outcome of an iteration in architecting based on deductive research as described in section 2 (see Fig .1). The iteration aims first to find out the suitable explaining constructs (e.g. Freirean dialogue as act of knowing) and second to structure them in a way that enable their effective use in context of initializing a program of workshops as act for liberation.

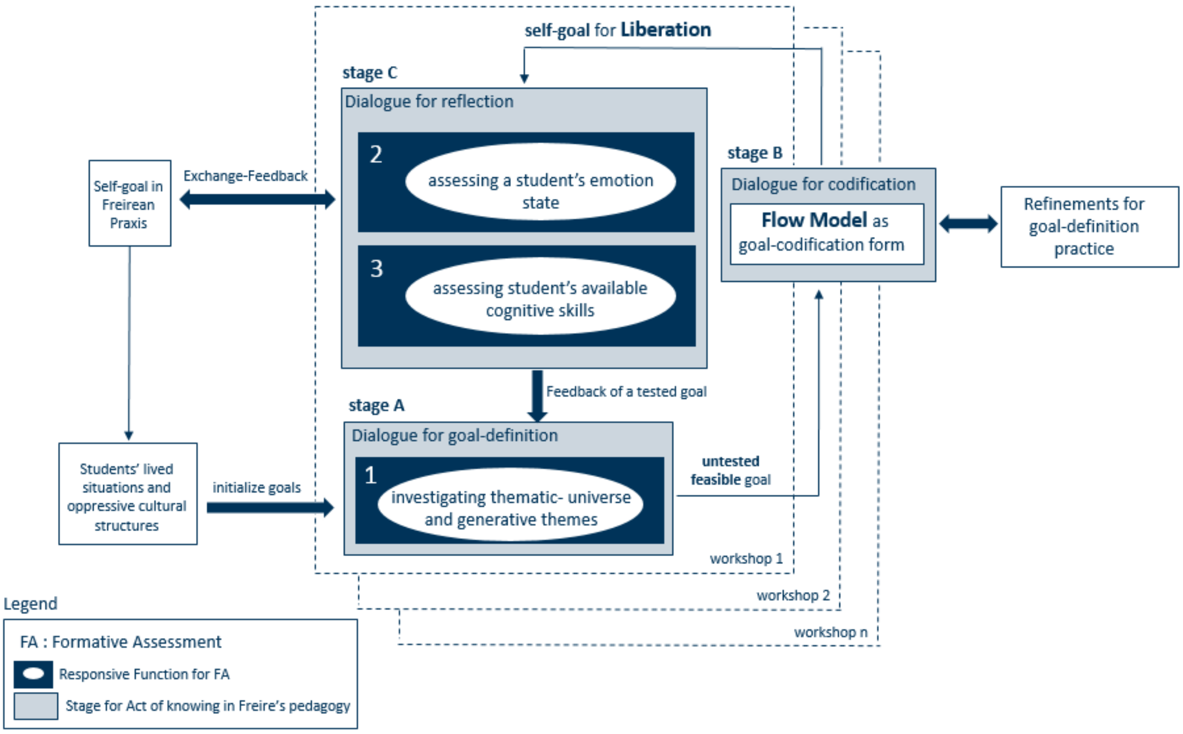

Figure 3. Architecture as Guidelines for Co-humanists. It helps trainers and students integrate Freirean dialogues and flow-model in initializing and further designing a program of workshops.

To facilitate the presentation of how the 3 functions in Fig.3 can be effectively used, it is required firstly to clarify the meaning of five Freirean concepts. The Freirean concepts are:

- **Dialogue**: to enter dialogue presupposes equality amongst participants. Each must trust the others; there must be mutual respect and love (care and commitment). Each one must question what he or she knows and realize that through dialogue existing thoughts will change and new knowledge will be created. [15]
- **Praxis (Action/Reflection)** : It is not enough for participants to come together in dialogue to gain knowledge of their social reality. They must act together upon their environment to reflect critically upon their reality and so transform it through further action and critical reflection [15]

- **Generative Themes**: According to Paulo Freire, an epoch "is characterized by a complex of ideas, concepts, hopes, doubts, values and challenges in dialectical interaction with their opposites striving towards their fulfilment". The concrete representation of these constitute the themes of the epoch. For example, we may say that in our society some of these themes would include the power of bureaucratic control or the social exclusion of the elderly and disabled. In social analysis these themes may be discovered in a concrete representation in which the opposite theme is also revealed (i.e., each theme interacts with its opposite) [15].
- **Conscientization**: The process of developing a critical awareness of one's social reality through reflection and action. Action is fundamental because it is the process of changing the reality. Paulo Freire says that we all acquire social myths which have a dominant tendency, and so learning is a critical process which depends upon uncovering real problems and actual needs [15].
- **Codification:** This is a way of gathering information in order to build up a picture (codify) around real situations and real people. Decodification is a process whereby the people in a group begin to identify with aspects of the situation until they feel themselves to be in the situation and so able to reflect critically upon its various aspects, thus gathering understanding. It is like a photographer bringing a picture into focus. [15]

The three responsive functions for FA, as seen in Fig.3, are about facilitating the **liberation-oriented** defining and assessing of student self-goals. As seen in Fig. 3, the initial goals in each workshop iteration are collected from student's lived situations. The initial goals then are used as input to investigate the thematic-universe and the generative themes via function 1. Function 1's outcome is an **untested feasible** goal (for more details see untested-feasibility in [7]), that is used as input for a goal-definition practice based on flow-model. The practice is driven by refinements (examples are described in table 2) structured for improving the goal-definition as seen in Fig.3. The practice is conducted via Freirean dialogue for codification. As codification outcome, a student self-goal for liberation is produced in an **accessible an assessable** form to facilitate the responsive functions 2, and 3. Functions 2 and 3 are conducted via Freirean dialogue for formative assessment and reflection. As outcome of Function 2 and 3, there is two types of feedbacks; namely, the exchange-feedback that is very essential to enable a student implement his/her self-goal by in Freirean praxis, and the feedback of a tested goal that is fed to refine the function 1 and started a new workshop iteration.

To sum up, the responsive function 1 is conducted in stage A, (i.e. via Freirean dialogue for goal-definition, whereas the responsive function 2 and 3 are conducted in stage C (i.e. via Freirean dialogue for reflection). Between stage A and C, the stage B. In stage B, the flow-model is used as tool for goal-codification (via Freirean dialogue for codification). The iterative conducting of Freirean dialogues in stages A, B and C enables the liberation-oriented defining and assessing of student self-goals. This, in turn, helps trainers and students co-initialize and further co-design the suitable adult literacy program of workshops as act for liberation. In next section, we present how the flow-model is used as guideline form for goal-codification.

4.2 Flow-model as Goal-Codification Form

As guideline to codify the challenges and skills of a goal in a measurable style for better analysing and understanding the disengagement problems, we select the flow model (see Fig. 4). The flow model is explained by Csikszentmihaly's research outcomes as seen in Fig. 4 [1, page 72]. Csikszentmihaly's research is about measuring and analysing optimal everyday experiences (e.g. flow experiences) and their negative/positive emotional states in everyday life. The student everyday experiences (lived situations) play important role in defining the initial goal and practicing Freire's dialogic pedagogy. Both Freire's dialogic pedagogy and Csikszentmihaly's flow theory are about understanding everyday experiences, therefore an architectural decision was taken by us, in context of our research based architecting activities, to integrate them in our method as seen in Fig. 3.

Fig. 4 summarizes Csikszentmihaly's research outcomes in the map of everyday experience. The map has eight emotional/mental states, namely; control, boredom, apathy, worry, anxiety, arousal, and flow. A concrete example of everyday activity is car driving. When driving a car, people are most often in control state, provided they are not stuck in a traffic jam, in which case they would be in boredom or apathy; or if there is no sudden snowstorm, in which case they are likely to be in anxiety. The car driving example is used by us, during the Freirean dialogues, as metaphor to explain to the student how we use the flow model in our offers to support him/her in finding individual growth paths (roadmap) for skills mastery. The Freirean dialogues for codification (stage 2 in Fig. 3) are structured not only around student everyday experiences but also around the flow model and its two dimensions and central point as seen in Fig. 4. The flow model, its dimensions and its central point are explained and interpreted in [1] for many everyday experience contexts, but the context of everyday "programming task" experiences is not included, therefore for this context we summarize our interpretation as it follows.

The central point represents the weekly average level of challenges and skills in "programming task" experiences. This weekly average level is individual and personal. That means it must be estimated for each student separately because it depends on student personal attributes like everyday habits, background, and prior skills. The estimation is based on scoring criteria in experience sampling method [17]. In brief, it is about estimating the weekly average of student performance score in knowing-how to align (or to balance) challenges and skills based on colleting the evidences, during a week, about his/her everyday "learning" or "programming task" experiences. The closer a student's score to this central point, the more average his/her mood tend to be - neither positive no negative. However, as his/her score moves away from this central point, distinct states of mind begin to emerge depending on the ratio of challenges and skills. Basically, the more a person feels skilled, the more her moods will improve; while the more challenges that are present "for immediate goal or task at hand", the more her attention will become focused and concentrated. As we would expect, optimal experience is represented by the flow "state" in Fig. 4 where both challenges and skills lie above the average level - at such moments of optimal experience one is both happy and focused.

The challenges dimension represents "different combinations of opportunities to act". We provide the opportunities as outcome of our "goal-codification or definition" practices. They are provided in many codification or definition forms (For forms in concrete examples, see please table 2). The forms facilitate a very high ceiling of student challenges for skills mastery improvement. In these forms, the student challenges are not only clear and actionable but also easy to be selected and combined by the student in context of activating him/her as owner of his/her learning. Moreover, the challenges dimension has scale levels between minimum level and maximum level, where the average level, between the maximum and the minimum, refers to the central point (see Fig. 4).

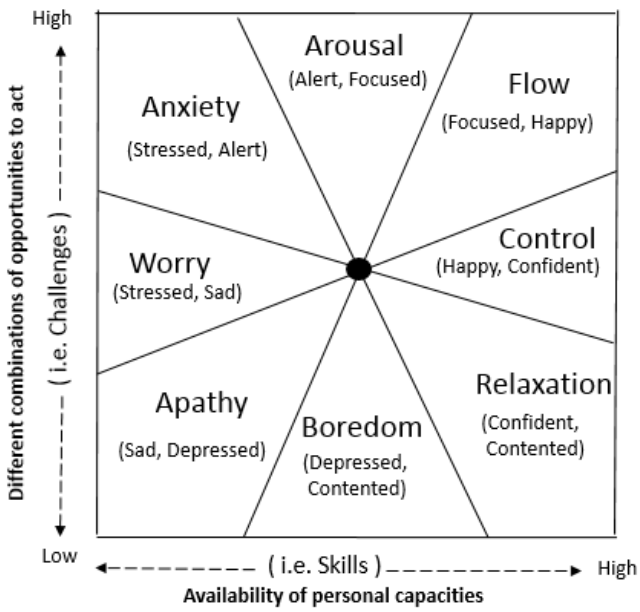

Figure. 4. Flow Model; the Map of Everyday Experience. When people perceive themselves to be above their own personal average level of challenges and skills, they experience flow. The opposite is the state of apathy, where both challenges and skills are low. Other combinations of challenges and skills produce feelings of worry, anxiety, and arousal (when challenges outweigh skills), or control, relaxation and boredom (when skills outweigh challenges). Some of the other prominent emotions typical of each "state" are indicated in parentheses [1, page 72].

The skills dimension represents "availability of personal capacities". The availability is about timing and attention resource in one's psychological capital. Psychological capital is a metaphor to simplify explanation: It is useful to think of **enjoyment** as the psychological equivalent of building capital, and of **pleasure** as the equivalent of its consumption. As used in economics the term "capital" would be defined as follows: Capital refers to resources withheld from immediate consumption in the expectation of greater future returns [1, page 76]. To put it differently, “enjoyable activity” builds one’s psychological capital, whereas “pleasant activity” consumes the psychological capital. *But what exactly, is the resource at psychological level that will be “consumed” during a “pleasant activity”, however in an “enjoyable activity” it will be withdrawn from consumption?* The answer to last question is “**attention resource**”. Attention is the brain’s capacity to process information and to direct action. It is a limited resource, because we cannot process more than a few bits of information at any single moment, and thus we can only be aware of a tiny fraction of what is going on inside us or around us. **Attention is psychic energy**, and like physical energy, unless we allocate some part of it to the task at hand, no work gets done. The skills dimension has, like challenges dimension, scale levels with minimum level, maximum level and average level that refers to the central point as seen in Fig. 4.

To sum up, the model’s 8 emotional states in Fig. 4, provide a serviceable compass for finding one’s way through the thickets of the emotional life. The direction toward which the model clearly points is flow. Beside flow there are two states in Fig. 4 associated with positive emotions. The first is the one labeled “arousal” where due to slightly higher challenges a person must concentrate but does not feel quite at ease. In this situation, if one wants to enter flow, the level of skills must be improved. Because flow state is attractive, a person in arousal is likely to be motivated to reach that state, and thus learns and grows to accomplish that. The other positive state is “control” where skills slightly outweigh challenges. This “control” state does not require high concentration and therefore one does not operate at 100 percent capacity, and so it is not as enjoyable as flow. However, it is relatively easy to move from this state or condition into flow, by somewhat higher challenges. Arousal and control are both states that lead to engagement in ***learning*** because they ***stimulate a student or a person to develop higher complexity of skills***. Via our support as trainers, we aim to create conditions and situations that help student experience arousal, control and flow. To put differently to enable creating situations where student’s skills are engaged by progressively higher challenges of learning.

In contrast to arousal, control and flow, the states of boredom, apathy, worry, and anxiety lead to disengagement in learning because they produce situations where student’s skills are not engaged by progressively higher challenges to stimulate a student’s critical thinking.

To sum up, in Fig.3, we present our practice-method that has three iterative stages A, B and C. The stages enable Freirean Dialogues for goal-definition, for goal-codification, and for reflection. Beside Freirean Dialogues the method facilitates the effective use of Flow-model and three responsive functions for FA as essential elements to initialize and design a literacy program of workshops for a group of oppressed students. Beside Freirean Dialogue and Flow-model as essential design elements, elements for how to use and integrate the flow-model in Freirean codification (i.e. stage b) are also clarified. Using and integrating Freirean dialogues and flow model are presented to clarify how our practice-method is conceptualized as helping tool for **co-humanists** (i.e. trainers and students). The tool helps them use **investigation driven iterative process** (see Fig 3) to initialize and design an adult literacy program of workshops for reading and writing software programs**.** The role “co-humanist” is a role to clarify how to use our mothed. It is designed by us as outcome of our interpretation of what Freire called a **humanist**-literacy campaign and **not humanitarian**-literacy campaign. In next section, we present in details what is the role “co-humanist”.

4.3 Co-humanist Role in our Practice-method.

According to Freire, his pedagogy is about planning an adult ***humanist*** literacy program, and ***not Humanitarian*** literacy program. To clarify the difference we prefer to present Freire’s voice and quotes here. The quotes [16] are:

- In the culture of silence the masses are “mute” that is, they are prohibited from creatively taking part in the transformations of their society and therefore prohibited from being. Even if they can occasionally read and write because they were “taught” in **humanitarian**-but not **humanist-** literacy campaigns, they are nevertheless alienated from the power responsible for their silence. Illiterates know they are concrete men (i.e. humans). They know that they do things. What they do not know in the culture of silence - in which they are ambiguous, dual beings - is that men’s actions as such are transforming, creative, and re-creative. Overcome by the myths of this culture of silence, including the myth of their own “natural inferiority,” they do not know that their action upon the world is also transforming. Prevented from having a “structural perception” of the facts involving them, they do not know that they cannot “have a voice,” that is, that they cannot exercise the right to participate consciously in the socio-historical transformation of their society, because their work does not belong to them.

To put in different way, in culture of silence the monologic relations of domination, manifesting in the mainstream educational system that Freire called, “the banking model of education”[7]), cannot produce reflective knowledge. In such model, the participants, can only participate in primitive non-reflective knowledge. They cannot be involved in knowing about their own knowledge. They can change the repertoire of their behaviors or activities as a result of this monologic education but they cannot fully master it because they are not involved in understanding the mastery of their non-reflective knowledge. Reflective knowledge requires true dialogues to enable reflective questions of “why”, “for whom”, “at whose expense”, and so on. Each time when a student’s question is not answered, or suppressed, or not taught to ask, the knowledge remains ***ontologically unhumanized*** and ***epistemologically*** **unreflective**. Any situation in which some individuals prevent others from engaging in the process of inquiry is one of violence. The means used are not important; to alienate human beings from their own decision-making is to change them into objects.

The co-humanist role enables at least two persons (educator and learner), to plan an adult ***humanist*** literacy program of workshops as act of knowing. It enables two knowing subjects and a knowable social reality (e.g. a codified existential situation) to meet in dialogue (in written or spoken forms). In dialogue, a synthesis is aimed to be achieved. A **synthesis** is between the educator’s **maximally** systematized knowing and the learner’s **minimally** systematized knowing. The educator’s role is to propose or pose problems about the codified existential situations in order to help the learners arrive at a more and more **critical view** of their reality [16].

The adult humanist literacy program as an act of knowing implies the existence of two interrelated contexts; The first is a concrete context (stage A in Fig.3). It is the real, concrete context of facts, the social reality (lived situations) in which a group of oppressed students/learners exist. The second is **theoretical** (stage C in Fig 3). It is the **context** of authentic dialogue between learners and educators as equally knowing subjects (i.e. co-humanists). As knowable object, the codification (stage B in Fig.3) mediates between the concrete and theoretical contexts (of reality). It mediates between the knowing subjects, educators and learners, who seek in dialogue to unveil knowledge for action and reflection. A set of co-humanist activities for action, reflection and codification is what consists the adult co-humanist literacy program. Developing Co-humanist literacy program for groups of adult oppressed students from countries in war (e.g. Syria or Ukraine) is planned in our future works. The future works and conclusion is presented in next section.

5 CONCLUSION AND FUTURE WORKS

To conclude, in this work-on-progress paper, we have conceptualized a practice-method as helping tool for trainers and students. The tool helps them use **investigation driven iterative process** (see Fig 3) to co-initialize and co-design an adult literacy program of workshops for reading and writing software programs (e.g. data analytics APIs). It helps them practice how to use and integrate Csikszentmihalyi’s flow-model [1] and Freirean Dialogues for **F**ormative **A**ssessment (FA). it enables three responsive functions for FA: namely 1) function for flow-model based diagnosing of student’s emotional states (e.g. boredom, apathy, worry, and anxiety) that hinder his/her engagement, 2) function for analysing student’s oppressive cultural structures/lived situations that hinder his/her engagement, and 3) function for assessing student’s available cognitive skills to facilitate matching balance between challenges and skills in goal-definition. The method enables also integrating the flow-model as goal-codification form. In this form, the trainers can codify the challenges and skills of a goal in a measurable style for better analysing and understanding the disengagement problems. Putting the method on action via co-developing (i.e. co-initializing and co-designing) an adult literacy program for a group of oppressed students from countries in war (e.g. Syria) is our next future tasks.

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